M. FADIL FAIZ

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PROFESSIONAL SUMMARY

Detail-oriented Computer Engineering Technology student with a strong technical foundation in backend web development, machine learning, and hardware automation. Proven capability in building structured RESTful APIs, engineering convolutional neural networks (CNN) for medical image classification, and developing NLP models. Active researcher handling embedded systems and motor driver architectures. Experienced in collaborative capstone project delivery and dedicated to building scalable software and intelligent autonomous systems.

EDUCATION

Politeknik Negeri Indramayu (Polindra) | Bachelor of Applied Science (D4) in Computer Engineering Technology | 2025 – Present

- Current Academic Standing: Second Semester Student.
- Core Requisites: Data Structures, Advanced Programming, Embedded Systems, and Applied Mathematics.
- Active Researcher at UKM Robotika Polindra (Hardware & Research Division).
- Successfully approved for a 9 SKS credit conversion from advanced external professional curriculum towards current degree tracks.

SMA Negeri 1 Bumiayu | Natural Sciences (Ilmu Pengetahuan Alam)

- Final Grade: 85.5
- Built a strong foundation in analytical thinking, mathematics, and exact sciences.

TECHNICAL SKILLS

Backend & Web: Node.js, FastAPI, FlaskAPI, JavaScript (ES6+), Python.

Machine Learning & Deep Learning: Convolutional Neural Networks (CNN), Natural Language Processing (NLP), TensorFlow, Keras.

Databases & Tools: MongoDB,MySQL, Git, Github

Hardware & IoT: ESP32 & Circuit Simulation (Proteus, EasyEDA).

KEY PROJECTS

SummaAI (Web-Based AI Productivity & Automation Tool) | Collaborative Capstone Project

- Co-developed a production-ready web platform tailored for business automation and document intelligence.
- Architected core modules for automated PDF summarization and natural language processing (NLP) to parse and extract actionable items seamlessly.

- Utilized SvelteKit for an interactive user interface alongside FastAPI and PostgreSQL for secure database records, asynchronous processing, and OCR engine binding.

Medical Anomaly & Kidney Disease CNN Classifier | Deep Learning Project

- Engineered a custom Convolutional Neural Network (CNN) architecture from scratch using TensorFlow/Keras to analyze medical CT Scan image modalities.
- Configured a 32-neuron initial convolution layer and tuned hyperparameters to reach a robust validation accuracy of 93%.
- Evaluated model precision, recall rates, and confusion matrices before saving and exporting the optimized binaries to an .h5 deployment format.

Roblox Web Sentiment Analysis | Machine Learning & NLP Project

- Built a Natural Language Processing (NLP) model to automatically analyze the sentiment of Roblox application reviews on the Google Playstore.
- Handled data preprocessing, tokenization, and text classification to determine user satisfaction metrics based on raw text input.

Automotive Catalog Marketplace (JDM Car Sell) | Full-Stack Website Project

- Built a secure, clean-architecture RESTful backend system utilizing Node.js, Express, and MongoDB Atlas for a Japanese Domestic Market (JDM) vehicle catalog.
- Structured secure environment variable layers, data validation parsing, and production error-handling mechanisms.

Embedded Soccer Robot Control Systems | UKM Robotika Research Project

- Programmed firmware on an ESP32 microcontroller to process incoming remote controller inputs via Bluetooth and handled custom multichannel PWM tracking logic.
- Designed industrial-grade wiring systems with BTS7960 high-current motor drivers to regulate motor acceleration, protect circuits, and deliver steady high-torque power.

CERTIFICATIONS & PROFESSIONAL TRAINING

Pijak x IBM SkillsBuild Scholarship: Artificial Intelligence Path | Present

- Successfully achieved the Mandatory Achievement 2 program milestone within the professional cloud and software developer learning track.
- Demonstrated mastery in advanced Object-Oriented Programming (OOP) in Python, data structures, and computer science fundamentals.

IDCamp x Indosat Scholarship: Certified Backend Developer (Dicoding) | January 2026

- Completed formal training specializing in building production-grade backend servers and network microservices.
- Acquired hands-on experience deploying APIs using the Hapi Framework, ensuring input sanitization, dynamic routing, and streamlined async request lifecycles.